Title of the Thesis

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Author Full Name

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Programme Name

Supervisor:
Prof. Supervisor Name, PhD,
Position, Department,
Iscte – Instituto Universitário de Lisboa

Co-Supervisor:
Prof. Co-Supervisor Name, PhD,
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Institution

Month, Year

To my family.

Acknowledgments

I would like to thank my supervisor, **Prof. Supervisor Name**, for the guidance and support throughout this work.

Resumo

Este trabalho aborda o problema de ...

Palavras-chave: palavra-chave 1, palavra-chave 2, palavra-chave 3.

Abstract

This thesis addresses the problem of ...

Keywords: keyword 1, keyword 2, keyword 3.

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CHAPTER 1

Introduction

This chapter introduces the addressed problem, motivates the work, states the research questions, summarises the contributions, and outlines the remainder of the document.

1.1. Motivation

1.2. Problem Statement

1.3. Research Questions

This work addresses the following research questions:

RQ1.

RQ2.

RQ3.

1.4. Objectives

The main objectives are:

-
-

1.5. Contributions

The main contributions are:

-
-

1.6. Document Structure

The remainder of this document is organised as follows. Chapter 2 reviews the relevant literature and prior work. Chapter 3 describes the proposed approach. Chapter 4 presents and discusses the experimental results. Chapter 5 summarises the work and outlines directions for future work.

CHAPTER 2

Literature Review

This chapter presents the review methodology used to identify relevant literature and synthesises the related work on three-dimensional left-ventricle reconstruction from cardiac MRI and myocardial wall-thickness measurement.

2.1. Review Methodology

The literature search followed a structured approach to ensure broad coverage and relevance. This section describes the search strategy, selection criteria, and results.

2.1.1. Search strategy

The search was conducted across three major bibliographic databases: Scopus, PubMed, and IEEE Xplore, supplemented by Google Scholar for citation chaining. The temporal scope covered publications from 2000 to 2026, prioritising work from the last decade while retaining foundational methods essential for theoretical context.

Search terms were grouped into four thematic categories and combined using Boolean operators. The first category covered anatomy and imaging terms ("cardiac MRI", "left ventricle", "short-axis", "SAX", "myocardium"). The second addressed reconstruction ("3D reconstruction", "mesh reconstruction", "shape model", "implicit surface", "signed distance function"). The third targeted learning methods ("deep learning", "graph neural network", "point cloud", "neural representation"). The fourth focused on measurement ("wall thickness", "myocardial thickness", "Laplace equation", "transmural").

The refined query applied consistently across databases was:
("cardiac MRI" OR "left ventricle" OR "short-axis") AND ("3D reconstruction" OR "statistical shape model" OR "signed distance function" OR "graph neural network" OR "implicit representation") AND ("wall thickness" OR "mesh" OR "deep learning")

2.1.2. Inclusion and exclusion criteria

Studies were included if they: (i) addressed three-dimensional reconstruction or shape modelling of the left ventricle, (ii) proposed or evaluated methods for myocardial wall-thickness measurement, (iii) developed geometric deep learning techniques applicable to anatomical meshes, or (iv) introduced benchmark cardiac MRI datasets used for LV analysis.

Studies were excluded if they: (i) focused exclusively on segmentation without reconstruction or shape analysis, (ii) addressed organs other than the heart without methodological generalisability, (iii) lacked peer review or sufficient methodological description,

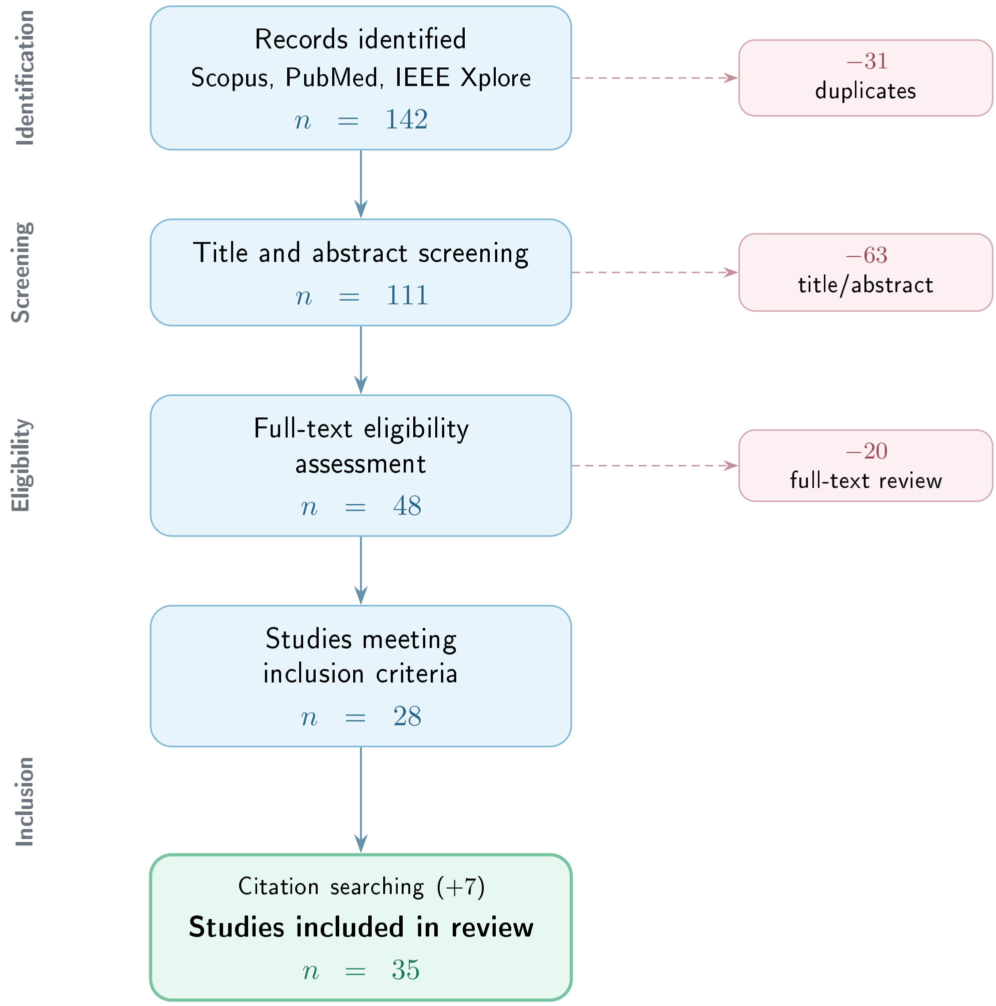

FIGURE 2.1. PRISMA flow diagram summarising the literature selection process.

or (iv) were published before 2000 unless they constituted foundational work directly relevant to the methods reviewed.

2.1.3. Results

The initial search retrieved 142 records across the three databases. After removing 31 duplicates, 111 unique records were screened by title and abstract. Of these, 48 were selected for full-text assessment. Following detailed review against the inclusion criteria, 28 studies were retained. An additional 7 studies were identified through backward citation searching, yielding a final corpus of 35 publications.

The selected studies span five thematic areas: cardiac MRI acquisition and segmentation, statistical shape models, implicit neural representations, graph neural networks for mesh processing, and wall-thickness measurement. The following section synthesises the key findings from this corpus.

2.2. Related Work

This section synthesises the reviewed literature, organised by thematic area.

2.2.1. Cardiac MRI segmentation and datasets

Short-axis (SAX) cine MRI is the standard protocol for left-ventricle (LV) evaluation, producing 2D slices perpendicular to the long axis with high in-plane resolution (1–2 mm) but large inter-slice gaps (6–10 mm) (Cerqueira et al., 2002; Petitjean & Dacher, 2011). This anisotropic acquisition creates the fundamental challenge that motivates 3D reconstruction: the ventricle must be inferred volumetrically from sparse planar observations. Petitjean and Dacher (2011) reviewed over 100 segmentation methods and concluded that the inter-slice gap remains the single largest source of error in volumetric cardiac analysis, with apex and base delineation accounting for the majority of inter-observer variability.

The American Heart Association (AHA) 17-segment model, introduced by Cerqueira et al. (2002), standardised the partition of the myocardium into basal, mid-cavity, and apical regions. This model is universally adopted for reporting wall motion, perfusion, and thickness, and serves as the reference frame for all regional analysis discussed in subsequent sections.

Segmentation of endocardial and epicardial contours is a prerequisite for any downstream 3D analysis. Ronneberger et al. (2015) proposed U-Net, an encoder–decoder architecture with skip connections that was originally designed for neuronal structure segmentation but rapidly became the de facto backbone for cardiac MRI analysis. The architecture trains effectively from small annotated datasets and produces pixel-level predictions that, when applied to SAX slices, yield endocardial and epicardial contours with sub-pixel accuracy.

The Automated Cardiac Diagnosis Challenge (ACDC), organised by Bernard et al. (2018), provided the first standardised cardiac segmentation benchmark. The dataset comprises 100 patients distributed across five clinical groups: normal, dilated cardiomyopathy (DCM), hypertrophic cardiomyopathy (HCM), abnormal right ventricle, and previous myocardial infarction. Each case includes expert delineations at end-diastole (ED) and end-systole (ES). The challenge reported that top-performing deep learning methods achieved mean Dice scores exceeding 0.93 for the LV cavity and 0.89 for the myocardium, demonstrating that automated segmentation had reached expert-level performance for standard acquisitions.

Isensee et al. (2021) introduced nnU-Net, a self-configuring framework that automatically adapts network topology, patch size, preprocessing, and post-processing to each dataset. Applied to the ACDC benchmark, nnU-Net matched or exceeded all prior challenge submissions without any manual architecture design. The framework has since become the standard baseline against which new cardiac segmentation methods are evaluated.

The Multi-Centre, Multi-Vendor and Multi-Disease (M&Ms) challenge, described by Campello et al. (2021), extended evaluation to the clinically critical question of domain generalisation. The dataset includes 345 subjects acquired on scanners from four different vendors (Siemens, General Electric, Philips, and Canon) across six clinical centres. The

[Placeholder] Example SAX slices from the ACDC dataset showing endocardial and epicardial contours for different pathological groups (normal, DCM, HCM) at end-diastole.

FIGURE 2.2. Representative short-axis slices from the ACDC dataset with expert segmentation contours across different cardiac pathologies (Bernard et al., 2018).

challenge demonstrated that while methods trained on a single vendor achieved high accuracy on that vendor, performance dropped significantly when tested on unseen vendors — with Dice reductions of 5–15 percentage points in some cases. This finding motivated the development of domain-adaptation techniques and highlighted the importance of multi-source training data.

For population-level analysis, the UK Biobank (Sudlow et al., 2015) provides the largest available resource, with cardiac MRI acquired for over 40 000 participants using a standardised 1.5T protocol. Bai et al. (2018) developed a fully convolutional pipeline to segment this cohort at scale, producing automated delineations that, after quality control, serve as the foundation for large-scale shape modelling and epidemiological studies.

2.2.2. Statistical shape models

Statistical Shape Models (SSMs) encode anatomical variability by aligning a training population to point-to-point correspondence and applying PCA to extract the dominant modes of variation (Cootes et al., 1995; Heimann & Meinzer, 2009). Cootes et al. (1995) introduced Point Distribution Models, where a training set of shapes — each represented by k corresponding landmarks — is rigidly aligned via Procrustes analysis, averaged into a mean shape $\bar{x}$, and decomposed via PCA into principal modes Φ. A new shape is generated as $x = \bar{x} + \Phi\, b$, where b are shape parameters constrained within $\pm 3\sigma$ of each eigenvalue to remain anatomically plausible. In their experiments on hand and face contours, Cootes et al. (1995) showed that the first 5–10 modes captured over 95% of population variance, demonstrating the efficiency of the linear representation.

Heimann and Meinzer (2009) surveyed over 100 studies applying SSMs to medical image segmentation and identified the cardiac domain as particularly well-suited because of the regularity of ventricular geometry and the availability of large training cohorts. Their review highlighted that SSM accuracy depends critically on three factors: the quality of inter-subject correspondence, the size and diversity of the training set, and the alignment method used.

Frangi et al. (2002) pioneered automatic construction of multi-object cardiac SSMs, proposing a non-rigid registration framework that establishes dense correspondence across subjects without manual landmarks. Applied to biventricular models, their approach captured coupled shape variation between the left and right ventricles, demonstrating that multi-structure models outperform single-structure ones for segmentation-by-fitting tasks.

Bai et al. (2015) constructed the most comprehensive cardiac atlas to date, using over 1000 high-resolution MRI scans from the UK Digital Heart Project. Their biventricular model captures both end-diastolic shape and wall motion through the cardiac cycle. The authors reported that the first 50 PCA modes explain approximately 90% of shape variance, and demonstrated that the atlas enables automatic segmentation of new subjects by fitting the model to image evidence with mean surface-to-surface errors below 1.5 mm.

de Marvao et al. (2014) applied the same UK Biobank-derived SSM to study myocardial hypertrophy and showed that atlas-based shape analysis detects subtle remodelling patterns that volumetric indices alone (such as LV mass) miss. Their study demonstrated substantially improved statistical power for detecting hypertrophic phenotypes when using shape modes compared to traditional clinical measurements, requiring fewer subjects to achieve equivalent sensitivity.

Medrano-Gracia et al. (2014) used a multi-ethnic cohort of over 1900 subjects to characterise normal LV shape variation. They reported that the dominant modes correspond to overall ventricular size, followed by eccentricity, apical shape, and regional wall curvature — confirming that cardiac shape cannot be adequately described by a single size parameter. Suinesiaputra et al. (2018) organised an international challenge to classify myocardial infarction from SSM-derived shape features, showing that machine learning classifiers trained on the first 20 shape modes achieved AUC values above 0.85 for distinguishing infarcted from healthy ventricles.

SSMs serve dual roles in reconstruction pipelines. First, they act as *geometric priors* that regularise shape recovery from sparse observations, constraining the solution to lie on or near the learned manifold. Second, they function as *synthetic data generators*: by sampling the latent space, one can produce thousands of anatomically plausible meshes for training data-hungry deep learning models without additional manual annotation (Heimann & Meinzer, 2009; Khalafvand et al., 2018). Khalafvand et al. (2018) demonstrated this second role by generating SSM-sampled LV geometries to train computational fluid dynamics simulations, showing that the synthetic population reproduced the haemodynamic statistics of real subjects.

2.2.3. Deep learning models for 3D shape reconstruction

Classical reconstruction methods interpolate between segmented SAX slices using splines or deformable surfaces, but cannot recover geometric detail between slices, particularly near the apex and base (Heimann & Meinzer, 2009; Petitjean & Dacher, 2011). Petitjean and Dacher (2011) noted that linear interpolation between contours produces visible staircase artefacts and fails to reproduce the smooth taper of the LV apex, while parametric models (prolate spheroids, B-spline surfaces) impose smoothness but cannot express patient-specific morphological detail. Modern approaches replace interpolation with learned shape recovery, falling into two broad families: template deformation and implicit field prediction.

Template deformation methods maintain a fixed mesh topology and learn per-vertex displacements conditioned on input observations. Wang et al. (2018) proposed Pixel2Mesh, which takes a single RGB image as input and iteratively refines an ellipsoidal template mesh through three stages of graph convolutional layers. At each stage, vertex features are enriched with image features sampled via learned projections, and the mesh is refined through unpooling. The authors reported Chamfer distances below 0.5 on the ShapeNet benchmark, demonstrating that graph-based deformation can recover complex topology from minimal input. Choi et al. (2020) extended this paradigm with Pose2Mesh, which recovers full 3D human body meshes from sparse 2D pose keypoints. Their cascaded coarse-to-fine architecture first predicts a coarse mesh from joint positions and then refines vertex locations with finer graph convolutions, achieving state-of-the-art results on the Human3.6M benchmark. The architectural principle — recovering dense 3D geometry from sparse 2D observations — is directly analogous to the cardiac problem of inferring a full LV mesh from a handful of 2D SAX contours.

In the cardiac domain, template deformation preserves the topological consistency required for cross-subject comparison and downstream analysis such as wall-thickness computation (Bai et al., 2015; Suinesiaputra et al., 2014). Beetz et al. (2021) applied point-cloud variational autoencoders to biventricular shapes from the UK Biobank, learning a compact latent space that enables generation of subpopulation-specific anatomy models. Their decoder produces meshes with fixed vertex correspondence, making population statistics directly computable from the generated samples.

Implicit Neural Representations (INRs) take a different approach. Rather than deforming a template, they encode geometry as a continuous function whose zero-level set defines the surface. Park et al. (2019) introduced DeepSDF, which parameterises the signed distance function (SDF) of an entire shape class with a single neural network conditioned on a per-shape latent code. Given partial or noisy point-cloud observations, their auto-decoder architecture optimises the latent code at test time to find the shape in the learned space that best explains the input. On the ShapeNet benchmark, DeepSDF achieved reconstruction quality comparable to voxel-based methods while using an order of magnitude fewer parameters, and naturally supports shape interpolation in latent space. Mescheder et al. (2019) proposed Occupancy Networks, which predict binary inside/outside occupancy rather than signed distance. Their encoder processes a point cloud or image to produce a latent vector, and the decoder evaluates occupancy at arbitrary 3D query points. Both DeepSDF and Occupancy Networks produce resolution-independent surfaces extracted via the Marching Cubes algorithm (Lorensen & Cline, 1987), which generates a watertight triangulated mesh from the zero-crossing of the implicit field on a regular grid.

Two techniques are critical for applying INRs to thin cardiac structures where high-frequency geometric detail must be preserved. Tancik et al. (2020) demonstrated that standard MLPs exhibit spectral bias — they preferentially learn low-frequency components — and proposed mapping input coordinates through random Fourier features before the network. In their experiments on 2D image regression and 3D shape reconstruction, Fourier encoding substantially reduced reconstruction error compared to raw coordinate input, with the improvement most pronounced for high-frequency details such as thin structures and sharp edges. Gropp et al. (2020) introduced Implicit Geometric Regularisation, showing that enforcing the eikonal constraint ($\|\nabla f\| = 1$) as a loss term during training produces valid distance fields without requiring ground-truth SDF values at every point. Their approach enables training from raw point clouds alone, which is particularly relevant when dense SDF supervision is expensive to compute.

For encoding unordered input observations such as SAX contour points, the PointNet architecture (Qi et al., 2017) provides a permutation-invariant encoder via shared MLPs followed by symmetric max-pooling. Qi et al. (2017) demonstrated that this simple design achieves classification accuracy of 89.2% on the ModelNet40 benchmark and produces compact global descriptors that capture the essential geometry of the input point set regardless of its size or ordering. In cardiac applications, each SAX slice contributes a set of contour points that can be processed by a PointNet-style encoder to produce a latent representation of the observed geometry.

2.2.4. Wall-thickness measurement techniques

Myocardial wall thickness is a primary clinical biomarker for cardiac remodelling. Grossman et al. (1974) demonstrated that wall thickness is a principal determinant of left-ventricular diastolic stiffness, with hypertrophied ventricles exhibiting markedly elevated stiffness proportional to end-diastolic wall thickness. More recently, Huelnhagen et al. (2017) showed at 7.0T MRI that wall thickness correlates with tissue transverse relaxation properties, confirming its value as a non-invasive surrogate for myocardial tissue characterisation. The AHA 17-segment model (Cerqueira et al., 2002) provides the standard framework for reporting regional thickness values.

The accuracy of wall-thickness measurement depends critically on the quality of the underlying surface reconstruction, as any smoothing or geometric artefact propagates directly into the thickness estimate. The literature describes five families of measurement methods, each with distinct trade-offs.

The simplest approach uses nearest-neighbour distance: for each endocardial vertex, a KD-tree identifies the closest epicardial point, yielding a measurement in $O(N \log M)$ time. While computationally trivial (under 1 ms for typical cardiac meshes), this method provides only a lower bound on the true transmural thickness because it ignores the anatomical direction of measurement — the nearest point may lie tangentially rather than across the wall.

Ray-based methods address this limitation by casting a ray from each endocardial vertex along its surface normal until it intersects the epicardium. The intersection is computed via the Möller–Trumbore algorithm, accelerated with a Bounding Volume Hierarchy (BVH) to $O(\log F)$ per query. This approach measures transmurally but is sensitive to normal noise: a slightly rotated normal can produce a spurious long or short ray. The Shape Diameter Function variant improves robustness by casting a cone of K rays (typically K=7) at a half-angle of 30° around the normal and reporting the median intersection distance, effectively filtering outlier hits caused by local mesh irregularities.

Volumetric methods operate on voxelised representations. The Euclidean Distance Transform (EDT) computes, for each myocardial voxel, the distance to the nearest boundary, and the sum of the endocardial and epicardial distance fields approximates local thickness ($t = D_{\text{endo}} + D_{\text{epi}}$). This estimate is exact for planar parallel boundaries and overestimates by approximately 5–10% for the curvatures typical of the LV mid-wall. The method is computationally efficient ($O(V)$ via the Saito–Toriwaki algorithm) but limited by voxel resolution.

PDE-based methods represent the gold standard for transmural thickness. Jones et al. (2000) introduced the Laplacian approach — originally for cortical thickness in brain imaging — which solves $\nabla^2\psi = 0$ with Dirichlet conditions (ψ=0 at the endocardium, ψ=1 at the epicardium). The uniqueness of the harmonic solution guarantees that the resulting streamlines never cross, producing a globally consistent transmural direction field. The local thickness is $t = 1/|\nabla\psi|$. While conceptually elegant, this formulation amplifies numerical errors where the gradient magnitude is small (thin walls, apex). Yezzi and Prince (2003) proposed an improvement: instead of dividing by $|\nabla\psi|$, they solve two advection equations ($\nabla\psi \cdot \nabla u = -1$ from the endocardium and $\nabla\psi \cdot \nabla v = -1$ from the epicardium), with the total thickness given directly as $t = u + v$. This Eulerian PDE formulation avoids the numerical instability of gradient division and produces identical results to the Laplacian method in smooth regions while being significantly more stable near the apex and in walls thinner than 4 mm.

Finally, mesh-based regularised correspondence methods combine an initial geometric estimate (normal projection or nearest neighbour) with iterative Laplacian smoothing on the mesh connectivity graph. At each iteration, each vertex's thickness is updated as a weighted average of its neighbours, anchored to the initial estimate. This produces spatially smooth thickness maps suitable for bullseye visualisation, at the cost of potentially attenuating real local thickness variation if the smoothing parameter is set too aggressively.

[Placeholder] Summary diagram of wall-thickness measurement families: (a) KD-tree nearest neighbour, (b) normal ray, (c) Laplace streamlines, (d) AHA-17 bullseye output.

FIGURE 2.3. Overview of wall-thickness measurement approaches, ranging from simple distance queries to PDE-based transmural streamlines, with results aggregated into the standardised AHA 17-segment bullseye.

2.2.5. Summary and research gap

The reviewed literature shows that each component of the LV analysis pipeline is well-studied in isolation: segmentation is effectively solved by deep learning, SSMs encode population-level shape priors, implicit representations provide continuous surface modelling, and multiple wall-thickness algorithms exist. However, few studies integrate these components into a single pipeline for the specific setting addressed in this thesis. The missing combination is a method that encodes sparse SAX contour points, decodes paired endocardial and epicardial surfaces as implicit signed distance fields, conditions the reconstruction on cardiac phase, guarantees positive wall thickness by construction, and trains jointly on real clinical data and SSM-derived synthetic samples. This gap motivates the approach described in the remaining chapters.

CHAPTER 3

Methodology

This chapter describes the methodology used to reconstruct 3D LV geometry from sparse short-axis contours and to measure myocardial wall thickness on the reconstructed model. The proposed pipeline, named CardioSDF, combines synthetic and real cardiac MRI-derived data, a phase-conditioned implicit neural representation, and a systematic comparison of local wall-thickness algorithms.

3.1. Overview

The complete thesis workflow can be read as a progression from sparse clinical observations to continuous three-dimensional geometry and then to quantitative wall-thickness summaries. It starts from segmented short-axis (SAX) cardiac MRI stacks or synthetic Statistical Shape Model (SSM) meshes. In both cases, the available geometry is converted into a common contour-based representation composed of endocardial and epicardial points sampled across SAX slices. These points are passed to a PointNet-style encoder that produces a compact latent code for the observed LV geometry. A signed-distance implicit decoder then predicts two coupled surfaces: the endocardium and the epicardium. The epicardial field is parameterised as a monotone offset from the endocardial field, which guarantees positive wall thickness by construction. The resulting surfaces are finally used to compute local thickness maps and regional AHA-17 summaries.

The methodological design is organised around four objectives. First, the input representation must be compatible with both synthetic and real cases. Second, the model must reconstruct smooth watertight LV surfaces from sparse 2D observations. Third, the decoder must produce a valid myocardial wall everywhere, not only at the observed slices. Fourth, the reconstructed geometry must support local wall-thickness measurements that can be visualised as surface colour maps and regional AHA-17 bullseye plots.

The chapter is therefore organised as a technical pipeline rather than as a single model description. Section 3.2 explains how synthetic and real data were converted into a shared cache representation. Section 3.3 describes the CardioSDF encoder and decoder, with particular attention to the monotone-epi parameterisation used to guarantee positive wall thickness. Section 3.4 presents the training and inference procedure, and Section 3.5 describes the wall-thickness algorithms used to evaluate local measurements on the reconstructed geometry.

Figure 3.1 summarises this end-to-end workflow and should be interpreted as an overview of the thesis methodology rather than as a figure belonging only to data preparation. The first panels illustrate the sparse SAX observation and contour extraction,

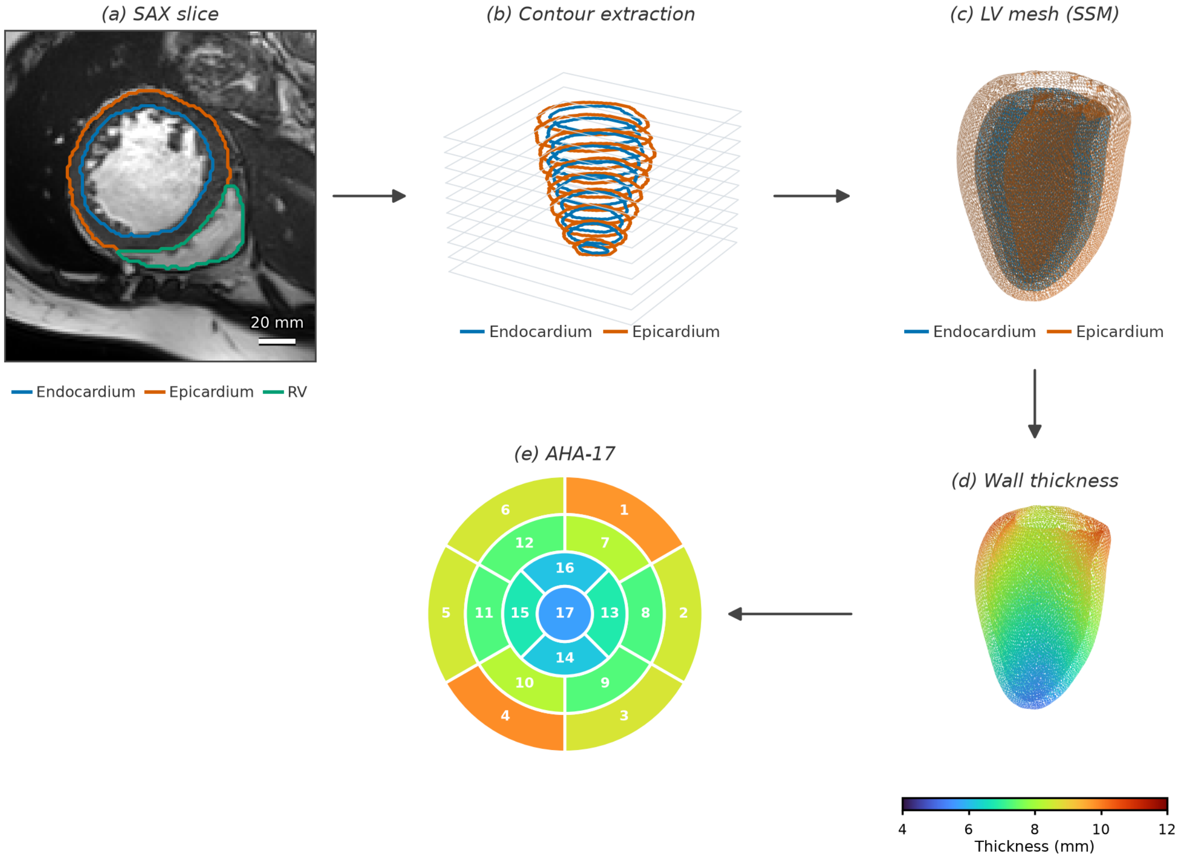

FIGURE 3.1. End-to-end overview of the thesis methodology. A representative SAX slice from ACDC patient002 illustrates the segmented clinical input and the corresponding endocardial, epicardial, and right-ventricular contours. The contour-extraction stage shows the sparse endocardial and epicardial observations used by CardioSDF, the mesh stage illustrates the reconstructed LV surface domain, and the final stages show local wall-thickness mapping and aggregation into the AHA-17 regional representation.

TABLE 3.1. High-level summary of the CardioSDF methodology.

Stage	Input / operation	Output
Data preparation	Synthetic SSM meshes and real ED/ES segmentations	Shared contour and query-point caches
Encoder	SAX contour points with tissue and phase labels	256-dimensional latent code z
Decoder	Latent code and Fourier-encoded query coordinates	Endocardial SDF, epicardial SDF, and bounded offset δ
Mesh extraction	Dense SDF samples on a 96^3 grid	Endocardial and epicardial triangular meshes
Thickness analysis	CardioSDF geometry and AHA orientation	Analytic thickness and 10-method local thickness maps

the mesh panel represents the reconstructed LV surface domain, and the lower panels show how the reconstructed geometry is converted into local and regional wall-thickness measurements.

3.2. Data Preparation

Three data streams were prepared for the thesis: a synthetic ED dataset generated from an SSM, a real ED dataset reconstructed directly from segmentation masks, and a real ES dataset processed with the same real-data pipeline. All datasets were converted into the same cache schema so that the training code could mix synthetic and real cases without changing the model interface.

The model is not trained directly from raw MR intensities. Each case is first converted into geometric supervision: endocardial and epicardial surfaces, short-axis contour rings, phase labels, and sampled query points. For real data, the starting point is a labelled segmentation volume. For synthetic data, the starting point is an SSM-generated mesh. The purpose of data preparation is to make these two sources indistinguishable to the model interface while preserving their different roles: synthetic data provides controlled shape variability, and real data provides acquisition- and segmentation-specific anatomy.

3.2.1. Problem visualisation and 3D slice inspection

Before designing the reconstruction pipeline, the segmented cardiac MRI stacks were visualised in three dimensions to make the sparsity of the input problem explicit. Each short-axis slice was placed at its physical through-plane position using the image spacing and affine information available in the NIfTI header. The LV cavity and myocardium labels were then overlaid on the slice planes, allowing the stack to be inspected as a set of parallel cross-sections rather than as an isolated sequence of two-dimensional images.

This visual inspection served as an exploratory quality-control step. It made it possible to verify whether the apical and basal slices were ordered correctly, whether the through-plane axis had been detected consistently after RAS+ canonicalisation, and whether the LV and myocardial labels matched the expected dataset convention. More importantly, it showed the central reconstruction challenge addressed in this thesis: only sparse contours are directly observed, while the final objective is a continuous three-dimensional endocardial and epicardial geometry. The 3D slice view therefore motivated the later conversion from labelled slices to contour rings and, finally, to implicit SDF supervision. The illustrative views in this section were generated from ACDC patient002, a dilated cardiomyopathy case with end-diastolic and end-systolic frames indicated in the accompanying metadata (Bernard et al., 2018).

3.2.2. Data sources and cardiac phases

The synthetic stream contains end-diastolic LV shapes sampled from the SSM. The real streams contain segmented cine MRI frames at end-diastole (ED) and end-systole (ES). These phases are not interchangeable. At ED, the LV cavity is near its maximum volume; at ES, contraction reduces the cavity size and changes the relative position of endocardial and epicardial contours. Therefore, the real-data pipeline keeps ED and ES as separate data streams and the model receives the cardiac phase as an explicit input feature.

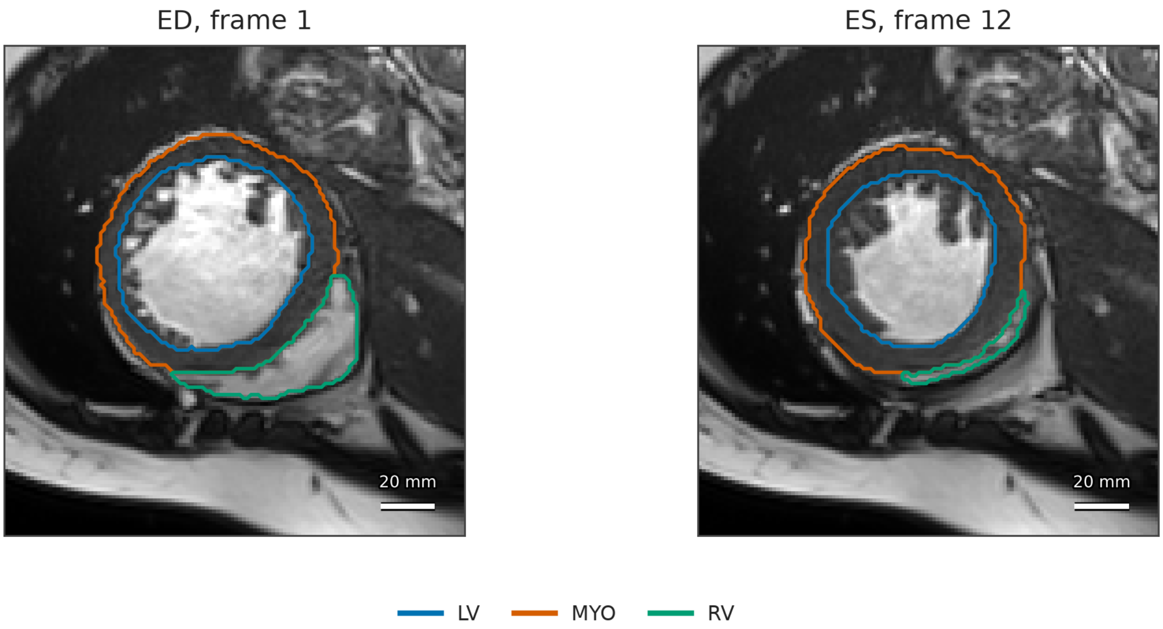

FIGURE 3.2. Representative SAX slices generated from ACDC patient002. The panels show ED frame 1 and ES frame 12 with contours for the LV cavity, myocardium, and right ventricle. This view was used to verify frame selection, label convention, and the visual separation between the LV cavity and myocardium before contour extraction.

For each real case, the segmentation labels provide the LV cavity and myocardium. These labels are converted into an endocardial mask and an epicardial mask, and then into surfaces and contours. For each synthetic case, the SSM provides the endocardial surface, and the epicardium is generated by a controlled normal offset. Once surfaces are available, both real and synthetic data are reduced to the same SAX-like observation format.

3.2.3. Synthetic end-diastolic dataset

The synthetic ED dataset was generated from the UK Digital Heart Project LV SSM, which provides a mean LV shape and 100 principal modes of variation (Bai et al., 2015). The SSM was used as a controlled source of anatomically plausible LV geometry, allowing the training set to include shape variation beyond the available real segmentations. In the notebook, the mean shape contained 22043 vertices, the PCA matrix had size 66129×100, and the eigenvalue vector contained 100 variances. A synthetic shape was created from the standard linear SSM model

$$X(b) = \bar{X} + \Phi b, \tag{3.1}$$

where $\bar{X}$ is the mean shape, Φ is the matrix of PCA modes, and b is the vector of sampled latent coefficients.

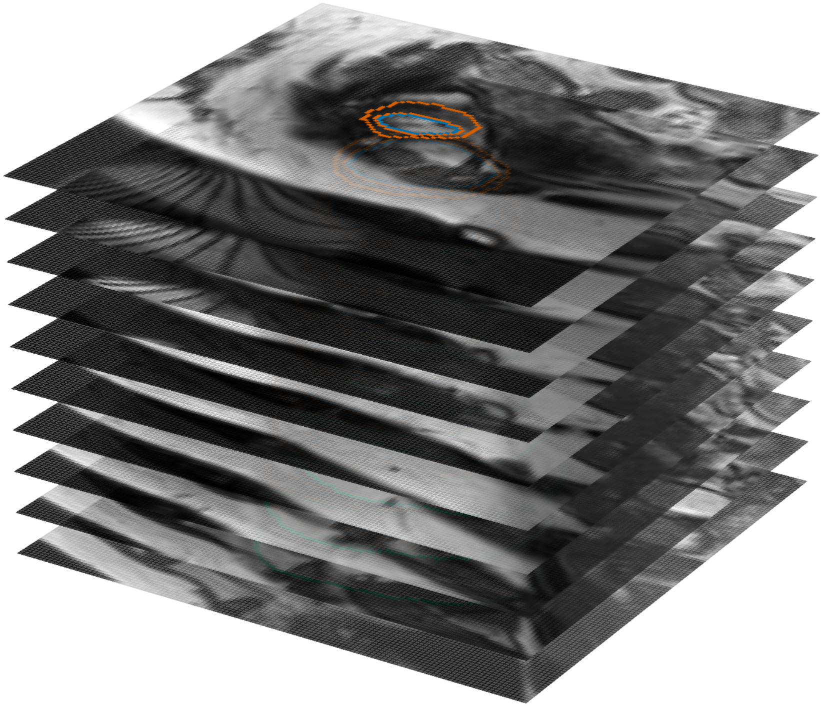

FIGURE 3.3. Problem visualisation through 3D inspection of the segmented ED SAX stack for ACDC patient002. The view places the slices according to their physical through-plane spacing and overlays the segmentation contours, highlighting the gap between sparse two-dimensional observations and the desired continuous LV surface geometry.

Latent coefficients were not sampled arbitrarily. Each accepted vector was constrained by a per-mode sigma clip and a global Mahalanobis bound,

$$\left|\frac{b_i}{\sigma_i}\right| \leq 3, \qquad \sum_i \left(\frac{b_i}{\sigma_i}\right)^2 \leq \chi^2_{0.99,d}, \tag{3.2}$$

where σ_i is the standard deviation associated with mode i and d is the number of active modes. This filtering prevents extreme samples from moving outside the plausible shape distribution. Each candidate mesh was further filtered using volume, surface area, sphericity, and apex/base plausibility checks.

The final synthetic generation run accepted 1300 meshes. The notebook also created pathology-like examples by exciting selected SSM modes to simulate normal, dilated cardiomyopathy, hypertrophic cardiomyopathy, and post-infarction geometries. Because the SSM describes the LV endocardium, a synthetic epicardial surface was produced by applying variable normal offsets, with a larger offset near the base and a smaller offset near

Table 3.2. Configuration of the synthetic ED dataset generation pipeline.

Component	Configuration
Shape source	UK Digital Heart Project LV SSM, mean shape plus 100 PCA modes
Accepted meshes	1300 synthetic ED meshes
Latent constraints	Mahalanobis 99% bound and per-mode 3σ clipping
Quality checks	Volume, surface area, sphericity, and apex/base plausibility
Pathology-like cases	Normal, DCM, HCM, and post-infarction mode excitations
Epicardium	Variable normal-offset surface, thinner near the apex
SAX extraction	10 SAX contour levels
Query cache	2048 near-surface and volumetric query points per case

the apex. Each mesh was then converted into 10 SAX contour levels, and the resulting contours were used to build training-compatible cache objects.

The synthetic pipeline served two purposes. First, it provided controlled geometric variation for training the encoder and decoder before relying entirely on real clinical segmentations. Second, it made it possible to generate pathology-like examples with known shape parameters, which is useful for testing whether the model can represent global LV remodelling. To avoid accepting unrealistic samples, the generation loop combined fast proxy metrics with periodic exact VTK audits. This ensured that the accepted meshes remained within physiological ranges for size and compactness while keeping the generation time manageable.

After mesh generation, each synthetic case was transformed into the same observation type available in real SAX MRI: a sparse set of endocardial and epicardial contour rings. Query points were sampled near the surfaces and in the surrounding volume, and inside/outside labels or SDF targets were cached for training supervision. This step is important because CardioSDF is not trained from complete meshes as direct input. The meshes define supervision, but the model input remains the sparse contour representation.

The synthetic epicardial surface was generated from the endocardial mesh by offsetting vertices along outward surface normals. The offset was larger near the base and smaller near the apex, reflecting the expected tapering of the LV wall. In simplified form, the epicardial vertex q_i associated with endocardial vertex p_i was computed as

$$q_i = p_i + d_i n_i, \tag{3.3}$$

where n_i is the outward normal and d_i is the local wall offset. The notebook used a base offset of approximately 10 mm, an apex offset of approximately 5 mm, and Gaussian perturbations to avoid perfectly uniform wall geometry. This synthetic epicardium is not intended to be a clinical truth, but it provides a consistent outer surface for pretraining and geometry supervision.

3.2.4. Real end-diastolic and end-systolic datasets

The real ED and ES datasets were prepared from ACDC, M&Ms, and M&Ms-2 segmentation masks (Bernard et al., 2018; Campello et al., 2021). The same core pipeline was used for both phases. Each NIfTI volume was first canonicalised to RAS+ orientation and resampled to an isotropic grid. The LV cavity label was used to define the endocardial mask, while the union of LV cavity and myocardium labels defined the epicardial mask:

$$M_{\text{endo}} = \mathbb{1}[L = L_{\text{LV}}], \qquad M_{\text{epi}} = \mathbb{1}[L \in \{L_{\text{LV}}, L_{\text{MYO}}\}], \tag{3.4}$$

where L is the segmentation label image, L_{LV} is the LV cavity label, and L_{MYO} is the myocardium label. Per-slice cleaning, hole filling, and three-dimensional largest-component filtering were applied before surface extraction.

For each cleaned mask, a smoothed signed-distance representation was converted to a mesh with marching cubes. The resulting surfaces were processed with Taubin smoothing, basal-plane capping, Laplacian fairing, and quadric decimation. The current notebook configuration uses 1.5 mm target spacing, 1.5 mm SDF smoothing, 25 Taubin iterations, 3 Laplacian fairing iterations, and a target of 4000 vertices. Anatomical quality checks were applied to reject cases with poor apical tapering, insufficient basal support, excessive centreline drift, irregular contours, invalid endocardium-epicardium ordering, implausible wall geometry, or unrealistic long-axis ratio. Real cases were split at patient level before augmentation to avoid data leakage between training, validation, and test sets.

The ED and ES pipelines deliberately avoid fitting the UK Biobank SSM to real segmentations. This choice is especially important for the ES phase, where an earlier SSM-fitting approach produced unrealistic systolic geometry. Instead, the real-data notebooks reconstruct surfaces directly from the observed segmentation masks. The SSM is therefore used for synthetic ED data generation, while the real ED and ES datasets preserve the patient-specific geometry present in the original segmentations.

The real-data preprocessing also standardises label conventions across sources. The datasets do not use identical encodings for the LV cavity, myocardium, and right ventricle. The notebooks therefore map each source to a common LV/myocardium representation before surface extraction. This harmonisation makes the downstream cache independent of the source dataset and prevents the training code from needing dataset-specific logic.

Surface extraction was treated as a reconstruction problem rather than a direct visualisation of voxel masks. A raw marching-cubes mesh from sparse SAX segmentations can contain staircase artefacts, disconnected fragments, and irregular basal closure. The real-data pipeline therefore smooths the volumetric boundary representation before surface extraction, keeps the largest anatomical component, applies volume-preserving Taubin smoothing, fits a basal plane, and decimates the final mesh. The basal cap is particularly important because SAX segmentations usually terminate at the valve plane; without a controlled cap, the reconstructed surface may contain a jagged or anatomically implausible base.

The ES cache uses the same processing logic as the ED cache, but it is built from end-systolic frames. This is not a minor implementation detail: the ES geometry has a smaller cavity and different wall configuration, and forcing a diastolic SSM to match systolic contours was found to produce unrealistic surfaces. Building ES meshes directly from real segmentations preserves the phase-specific anatomy needed by the phase-conditioned model.

3.2.5. Spatial normalisation and contour sampling

The source datasets differ in scanner orientation, voxel spacing, slice coverage, and label convention. To reduce this variability, real segmentation volumes are canonicalised to RAS+ orientation and resampled to an isotropic grid before surface extraction. The current real-data notebooks use a target spacing of 1.5 mm for mesh construction and an SDF smoothing scale of 1.5 mm. This normalisation is applied before marching cubes so that the extracted surfaces are defined in a consistent physical coordinate system rather than in arbitrary voxel units.

After surface extraction, all cases are converted into a common sparse SAX observation. Both the synthetic and real pipelines use 10 SAX contour levels; the real ED/ES caches store 60 contour points per ring. This fixed contour-level representation approximates the sparse through-plane sampling pattern of clinical SAX cine MRI while keeping a consistent input size for training. For supervision, 2048 query points are stored per case, sampled near the surfaces and in the surrounding volume. These query points allow the decoder to learn not only the observed contours but also the three-dimensional interior/exterior structure of the LV geometry.

3.2.6. Data augmentation

Data augmentation is applied to both synthetic and real cases at the contour-input level. The target meshes, surface samples, and cached query-point supervision remain fixed. In this sense, augmentation simulates uncertainty in the observed SAX contours rather than creating a different anatomical ground truth. This is important for the thesis because the real and synthetic data are not augmented by inventing new labels; instead, the same target geometry is observed through slightly perturbed contour inputs.

The augmentation operations include small in-plane translations, point jitter, rotation, scale perturbation, slice dropout, and contour-point dropout. These perturbations are phase-preserving: an ED case remains ED and an ES case remains ES. The goal is to make the encoder robust to realistic variability in contour extraction, slice placement, and segmentation noise while preserving the decoder's supervision target. Applying this strategy to both synthetic and real cases helps reduce the gap between clean SSM-derived examples and noisier clinical segmentations.

3.2.7. Training cache representation

All prepared cases were stored in a common representation composed of contour points, tissue labels, cardiac phase labels, mesh-derived supervision points, and cached SDF or

Table 3.3. Main operations in the real ED/ES mesh-preparation pipeline.

Operation	Purpose
RAS+ canonicalisation	Aligns scanner orientations across datasets
Label harmonisation	Converts ACDC, M&Ms, and M&Ms-2 to common LV/myocardium masks
Isotropic resampling	Reduces anisotropy before 3D surface extraction
Slice cleaning	Removes small 2D components and fills holes within SAX slices
3D component filtering	Keeps the main anatomical LV component
Marching cubes	Converts the volumetric boundary representation to triangular surfaces
Taubin smoothing and fairing	Reduces staircase artefacts while preserving global shape
Basal-plane cap	Produces a controlled closure at the valve-plane boundary
Quadric decimation	Reduces mesh size to a consistent target complexity
Anatomical prechecks	Rejects geometrically implausible or corrupted cases

Table 3.4. Summary of the datasets prepared for CardioSDF training and evaluation. Exact real-data sample counts should be filled after final cache export.

Data stream	Phase	Source	Main role
Synthetic SSM	ED	UK Digital Heart Project LV SSM	Anatomical shape variation and synthetic augmentation
Real MRI	ED	ACDC, M&Ms, M&Ms-2	Clinical end-diastolic geometry
Real MRI	ES	ACDC, M&Ms, M&Ms-2	Clinical end-systolic geometry and phase conditioning

occupancy queries. The real ED and ES caches use 10 SAX contour levels, 60 contour points per ring, and 2048 query points per case. This shared representation allows the final model to combine synthetic ED samples, real ED samples, and real ES samples in a single phase-conditioned training run.

The cache representation separates what the model observes from what the loss uses for supervision. The encoder observes only contour points, tissue identity, and phase. The decoder is supervised using surface samples, cached query points, and sign or SDF information derived from the training meshes. This distinction matches the target use case: at inference time, the model should reconstruct a full 3D LV geometry from sparse SAX contour observations rather than from a full 3D mesh.

3.3. CardioSDF Architecture

CardioSDF is composed of a PointNet-style encoder, Fourier positional encoding, and an implicit signed-distance decoder. The encoder maps sparse contour observations to a latent vector, while the decoder evaluates continuous SDF values at arbitrary 3D query

TABLE 3.5. Main fields in the shared training cache. Field names are descriptive rather than tied to a single file format.

Field	Meaning	Source	Used by
Contour points	SAX endocardial and epicardial points	Synthetic slicing or real masks	Encoder and anchor loss
Tissue label	Endocardium or epicardium identifier	Contour extraction	Encoder tissue pooling
Phase label	ED or ES cardiac phase	Dataset metadata	Phase-conditioned encoder input
Surface samples	Points on endocardial and epicardial meshes	Mesh extraction	Surface and normal losses
Query points	Near-surface and volumetric samples	Occupancy/SDF cache	SDF, sign, and exterior losses
Mesh metadata	Vertices, faces, scale, and orientation	Preprocessing pipeline	Evaluation and visualisation

points. The design follows the same general principle as point-cloud encoders and implicit shape decoders (Park et al., 2019; Qi et al., 2017), but it is specialised for paired LV surfaces and wall-thickness measurement. The model therefore separates the observed input representation from the resolution of the reconstructed mesh.

This architecture was chosen because the input and output have different geometric structure. The input is sparse, unordered, and slice-based, whereas the output is a continuous 3D surface. A point-cloud encoder is suitable for the input because it does not require a fixed grid or a fixed vertex correspondence. An implicit decoder is suitable for the output because it can be sampled at any resolution and converted into a mesh with marching cubes only at inference time. The monotone-epi coupling then adds the anatomical constraint needed for wall thickness: the epicardial field is not predicted independently, but is tied to the endocardial field through a positive offset.

3.3.1. Contour encoder

The encoder receives a point cloud whose features include spatial coordinates, tissue identity, and cardiac phase. A shared multilayer perceptron transforms each point independently into a feature vector. Separate max-pooling operations are then applied to endocardial and epicardial points, producing tissue-specific global descriptors. These descriptors are concatenated and projected to a 256-dimensional latent vector z. The use of max-pooling makes the encoder permutation-invariant, which is necessary because contour points have no fixed ordering across patients or slices.

Let $P = \{p_i\}_{i=1}^{N}$ denote the contour-point set for one case, where each point includes spatial coordinates, a tissue indicator, and a phase value. The shared point encoder computes

$$h_i = \phi_{\text{enc}}(p_i), \tag{3.5}$$

where ϕ_{enc} is the shared multilayer perceptron. Tissue-specific global descriptors are then formed with symmetric pooling,

$$z_{\text{endo}} = \max_{i \in \mathcal{E}} h_i, \qquad z_{\text{epi}} = \max_{i \in \mathcal{P}} h_i, \tag{3.6}$$

where $\mathcal{E}$ and $\mathcal{P}$ are the sets of endocardial and epicardial contour points. The final latent code is obtained by projecting the concatenated tissue descriptors,

$$z = W_z[z_{\text{endo}}, z_{\text{epi}}] + b_z. \tag{3.7}$$

Separating endocardial and epicardial pooling is important because the two contours encode different anatomical information. Endocardial points describe the LV cavity shape, while epicardial points determine the outer myocardial boundary. A single global pooling operation could mix these two roles and reduce the decoder's ability to distinguish cavity geometry from wall geometry. The per-tissue pooling step therefore gives the latent representation explicit access to both surfaces before they are fused.

3.3.2. Implicit decoder and monotone-epi coupling

Before query points are passed to the decoder, their coordinates are transformed with Fourier positional encoding. In the final training notebook, three frequency bands are used, producing a 21-dimensional encoded coordinate representation. This setting retains enough frequency content for LV-scale anatomy while reducing sub-voxel ringing and scattered marching-cubes artefacts. Fourier features reduce the spectral bias of coordinate MLPs and help the network represent geometric detail (Tancik et al., 2020). For a query coordinate $x \in \mathbb{R}^3$, the encoded feature vector is conceptually written as

$$\gamma(x) = [x, \sin(2^0\pi x), \cos(2^0\pi x), \ldots, \sin(2^{L-1}\pi x), \cos(2^{L-1}\pi x)], \tag{3.8}$$

with $L = 3$ frequency bands in the implemented model. The decoder is an eight-layer MLP with width 512 and a skip connection at the middle layer. It uses a softplus activation with high beta to preserve smooth implicit surfaces while avoiding the piecewise-linear creases associated with ReLU activations.

The decoder has two output heads. The first predicts the endocardial signed distance f_{endo}. The second predicts a bounded positive offset δ, interpreted as the local distance between the endocardial and epicardial fields. The offset is parameterised as

$$\delta(x, z) = \tau_{\min} + (\delta_{\text{cap}} - \tau_{\min})\sigma\left(g_\delta(x, z)\right), \tag{3.9}$$

where x is the query location, z is the latent code, g_δ is the raw decoder output, σ is the sigmoid function, $\tau_{\min}$ is the minimum normalised wall thickness, and δ_{cap} is the maximum allowed normalised offset. The epicardial field is then defined as

$$f_{\text{epi}}(x, z) = f_{\text{endo}}(x, z) - \delta(x, z). \tag{3.10}$$

Because Equation (3.9) enforces $\delta(x, z) \geq \tau_{\min}$, the epicardial surface is constrained to remain outside the endocardial surface in the model's signed-distance convention. In the

TABLE 3.6. Main CardioSDF architecture settings used in the final training notebook.

Component	Configuration
Input features	3D coordinates, tissue label, and cardiac phase
Encoder type	Shared MLP with endocardial and epicardial max-pooling
Latent dimension	256
Positional encoding	Fourier features with $L = 3$ frequency bands
Decoder	8-layer MLP, width 512, skip connection at layer 4
Activation	Softplus with high beta
Output heads	Endocardial SDF and bounded wall-thickness offset
Thickness range	$\tau_{\min} = 0.05$, $\delta_{\text{cap}} = 0.45$ in normalised units
Inference grid	96^3 query grid for marching cubes

final notebook configuration, $\tau_{\min} = 0.05$ and $\delta_{\text{cap}} = 0.45$ in normalised units, corresponding approximately to 1.25–11.25 mm under the 25 mm scale used during training.

3.4. Training and Inference

This section describes how CardioSDF is trained from the shared cache representation and how a surface mesh and wall-thickness map are extracted at inference time. Training is multi-objective because the model must learn both a geometric SDF and an anatomically valid relationship between the endocardium and epicardium.

3.4.1. Training data mix and augmentation

The final training configuration uses a combined ED and ES mode with phase conditioning. Synthetic ED, real ED, and real ES samples are all augmented at the contour-input level while their target meshes and cached supervision remain unchanged. The optimiser is AdamW with a learning rate of 2×10^{-5} and weight decay of 5×10^{-4}. Training uses mixed precision, gradient clipping, cosine annealing, early stopping, and gradient accumulation. On the Kaggle run described in the notebook, the model is parallelised across two NVIDIA T4 GPUs with PyTorch DataParallel.

The mixed dataset was chosen to balance coverage and realism. Synthetic ED data provides controlled shape variability and supports geometric augmentation, but it does not capture all segmentation artefacts present in real clinical data. Real ED and ES samples provide patient-specific anatomy from multiple sources and allow the model to learn the effect of cardiac phase. To avoid leaking patient identity across splits, real cases are split at patient level before any augmentation is applied. Augmentation is then performed on the contour observation, not on the anatomical target, so real segmentations remain tied to the observed patient anatomy.

The augmentation operations perturb only the encoder input. The cached mesh and query-point supervision remain fixed. This design simulates realistic contour uncertainty, such as small in-plane translations, jitter, missing slices, scale variation, rotation, or

Table 3.7. Purpose of the main CardioSDF training-loss terms.

Term	Purpose	Effect on reconstruction
$\mathcal{L}_{\text{surf}}$	Fits zero-level sets to surface samples	Places endocardial and epicardial surfaces on the target meshes
$\mathcal{L}_{\text{eik}}$	Encourages $\|\nabla f\| = 1$	Regularises the decoder as a signed-distance field
$\mathcal{L}_{\text{off}}$	Repels off-surface points from zero	Reduces spurious zero crossings away from the myocardium
$\mathcal{L}_{\text{normal}}$	Aligns predicted and target normals	Improves local surface orientation
$\mathcal{L}_{\text{WT}}$	Enforces the wall-thickness floor	Discourages collapse toward an unrealistically thin wall
$\mathcal{L}_{\text{L1}}$	Supervises cached query-point distances	Provides dense volumetric SDF information
$\mathcal{L}_{\text{anchor}}$	Anchors input contour points to the surface	Preserves agreement with observed SAX contours
Sign terms	Enforce inside/outside consistency	Stabilise the relative position of endocardium and epicardium
Bbox term	Pushes points outside the contour box outward	Suppresses exterior false surfaces

point dropout, without changing the target surface. In practical terms, the encoder is encouraged to be robust to imperfect sparse observations while the decoder is still supervised against a consistent 3D geometry.

3.4.2. Loss function

The loss function combines surface fitting, eikonal regularisation, off-surface repulsion, normal consistency, wall-thickness floor enforcement, cached SDF supervision, contour anchoring, sign-consistency penalties, and outside-bounding-box constraints. These terms serve complementary purposes: surface terms align the zero-level sets with the target meshes, eikonal and normal terms regularise the SDF geometry, contour anchoring preserves agreement with the observed SAX slices, and sign constraints prevent collapse of the myocardial wall.

The total training objective can be written as

$$\begin{aligned}\mathcal{L}_{\text{total}} = {} & \lambda_{\text{surf}}\mathcal{L}_{\text{surf}} + \lambda_{\text{eik}}\mathcal{L}_{\text{eik}} + \lambda_{\text{off}}\mathcal{L}_{\text{off}} + \lambda_{\text{normal}}\mathcal{L}_{\text{normal}} \\ & + \lambda_{\text{WT}}\mathcal{L}_{\text{WT}} + \lambda_{\text{L1}}\mathcal{L}_{\text{L1}} + \lambda_{\text{anchor}}\mathcal{L}_{\text{anchor}} + \lambda_{\text{sign}}\mathcal{L}_{\text{sign}} \\ & + \lambda_{\text{extsign}}\mathcal{L}_{\text{extsign}} + \lambda_{\text{bbox}}\mathcal{L}_{\text{bbox}} + \lambda_z\|z\|_2^2. \end{aligned} \tag{3.11}$$

The eikonal term is computed with automatic differentiation outside mixed precision to avoid numerical instability in the gradient calculation. The wall-thickness term complements the architectural lower bound on δ by penalising any residual tendency toward the minimum offset during training. The eikonal term follows the standard SDF regularisation principle that $\|\nabla f\|$ should be close to one for a valid distance field (Gropp et al., 2020).

Table 3.8. Training configuration used for CardioSDF.

Setting	Value
Training mode	Combined ED and ES with phase conditioning
Augmentation	Contour-input perturbations for synthetic and real cases
Optimiser	AdamW
Learning rate	2×10^{-5}
Weight decay	5×10^{-4}
Maximum epochs	400
Batch size	8, with gradient accumulation
Gradient clipping	1.0
Precision	Automatic mixed precision
Hardware	2 NVIDIA T4 GPUs with DataParallel
Inference grid	96^3

3.4.3. Optimisation and hardware

The final training setup uses AdamW with learning rate 2×10^{-5} and weight decay 5×10^{-4}. A cosine annealing schedule controls the learning rate, and early stopping prevents unnecessary training after validation stagnation. The notebook uses automatic mixed precision to reduce memory usage, gradient clipping at 1.0 to stabilise updates, and gradient accumulation to increase the effective batch size. The maximum number of epochs is 400, although early stopping can terminate training earlier.

The model was trained in a Kaggle environment with two NVIDIA T4 GPUs using PyTorch DataParallel. Spectral normalisation was disabled in the final notebook because of memory pressure, and a simpler weight-clipping strategy was used for lightweight Lipschitz control. These decisions reflect the practical constraint that the decoder is evaluated at many query points during training, making GPU memory a limiting factor.

3.4.4. Inference and analytic wall-thickness extraction

At inference time, the model encodes the input contour points into z and evaluates the decoder on a uniform 96^3 grid. Marching cubes extracts the zero-level sets of f_{endo} and f_{epi} (Lorensen & Cline, 1987). No mesh repair, cap synthesis, or PyMeshFix post-processing is applied to the predicted model outputs. Local analytic wall thickness is obtained by evaluating $\delta(x, z)$ at endocardial surface vertices and converting the result from normalised units to millimetres,

$$t_{\text{CardioSDF}}(x) = s\,\delta(x, z), \tag{3.12}$$

where s is the spatial scale used to map normalised coordinates back to millimetres. This analytic estimate is different from a post-hoc distance between two meshes: it is produced directly by the decoder parameterisation. The 10 additional wall-thickness algorithms described in Section 3.5 are therefore used as comparison methods on the same geometry, not as the only source of thickness information.

Table 3.9. Wall-thickness methods evaluated on the CardioSDF-generated LV geometry.

#	Method	Category	Main measurement principle
1	KD-tree nearest neighbour	Distance	Closest epicardial point for each endocardial vertex
2	Symmetric KD-tree	Distance	Bidirectional nearest-neighbour consistency
3	Normal rays	Ray	Intersection of endocardial normal rays with the epicardium
4	EDT boundary sum	Volumetric	Sum of endocardial and epicardial distance transforms
5	EDT medial axis	Volumetric	Thickness inferred from medial-axis distance ridges
6	Laplace field	PDE	Transmural field lines from a harmonic potential
7	Geodesic Dijkstra	Graph	Shortest paths through a 26-neighbour myocardial graph
8	SDF cone rays	Ray	Median of multiple cone rays around the surface normal
9	Regularised correspondence	Mesh	Smoothed correspondence field over the surface graph
10	Yezzi–Prince	PDE	Eulerian transport equations along the transmural field

3.5. Wall-Thickness Evaluation Protocol

The wall-thickness evaluation is performed on the CardioSDF-generated geometry. The segmentation volume is used to extract contour rings and to orient the AHA regional coordinate system, but the measured surfaces are the endocardial and epicardial meshes produced by the trained model. This distinction is important: the evaluation tests thickness estimation on the reconstructed implicit model, not directly on raw segmentation meshes.

3.5.1. Thickness methods

Ten wall-thickness methods are applied to the same reconstructed geometry: KD-tree nearest neighbour, symmetric KD-tree, normal rays, Euclidean Distance Transform boundary sum, Euclidean Distance Transform medial axis, Laplace field, geodesic Dijkstra, SDF cone rays, regularised correspondence, and the Yezzi–Prince Eulerian PDE method. Each method returns local thickness values in millimetres at endocardial vertices or projected endocardial locations. The outputs are summarised with mean, median, standard deviation, fifth percentile, ninety-fifth percentile, finite-value fraction, and runtime, and are visualised as 3D colour maps and AHA-17 bullseye plots.

The methods cover geometric distances, ray intersections, volumetric distance fields, graph distances, mesh regularisation, and PDE formulations. The nearest-neighbour baseline assigns each endocardial vertex p_i the distance to the closest epicardial vertex q_j,

$$t_i^{\mathrm{KD}} = \min_j \|p_i - q_j\|_2. \tag{3.13}$$

This is fast and easy to reproduce, but it does not guarantee that the closest point lies in the anatomical transmural direction.

The Laplace method instead defines a smooth scalar field ψ over the myocardium by solving

$$\nabla^2 \psi = 0, \qquad \psi|_{\mathrm{endo}} = 0, \qquad \psi|_{\mathrm{epi}} = 1. \tag{3.14}$$

The local wall thickness is then estimated from the gradient magnitude along the transmural field. This family of methods is attractive because harmonic field lines do not cross, but numerical errors can appear where the wall is thin or where the field gradient is small (Jones et al., 2000). The Yezzi–Prince approach addresses this by solving Eulerian transport equations along the same transmural field and combining the endocardial and epicardial travel times (Yezzi & Prince, 2003). In simplified form,

$$\nabla\psi \cdot \nabla u = -1, \qquad t = u + v, \tag{3.15}$$

where u and v are the propagation distances from opposite boundaries.

The cone-ray method uses the endocardial surface normal but reduces sensitivity to a single noisy ray. For each endocardial point, several rays are cast inside a cone around the normal direction, and the median valid hit distance is used:

$$t_i^{\mathrm{cone}} = \mathrm{median}_{k=1}^{K}\, d_{ik}, \qquad K = 7. \tag{3.16}$$

In the notebook implementation the cone half-angle is 30°. This makes the estimate more robust than a single normal ray while still preserving an approximately transmural interpretation.

3.5.2. Regional AHA-17 mapping

Local wall-thickness values are also aggregated into the AHA 17-segment model (Cerqueira et al., 2002). This requires identifying the LV long-axis direction, distinguishing base from apex, and establishing a septal reference direction. The wall-thickness notebook therefore includes an anatomical QA step that visualises base/apex orientation and septal alignment before constructing bullseye plots.

The AHA aggregation does not replace per-vertex thickness maps. Instead, it provides a clinically familiar regional summary of the same local measurements. For each method, endocardial vertices are assigned to basal, mid-cavity, apical, or apical-cap regions according to their normalised long-axis position, and then subdivided angularly around the LV centreline. Segment statistics are reported using robust summaries such as the mean and percentiles so that local outliers do not dominate the regional interpretation.

This evaluation should be interpreted as an algorithmic comparison and reproducibility study rather than as clinical validation. Unless expert manual wall-thickness measurements are added, the thesis can conclude that CardioSDF provides an objective and locally resolved computational framework, but it should not claim replacement of clinical measurement protocols.

CHAPTER 4

Results and Evaluation

This chapter presents the experimental evaluation of the approach described in Chapter 3.

4.1. Experimental Setup

4.1.1. Datasets

4.1.2. Metrics

4.1.3. Baselines

4.2. Results

4.3. Discussion

4.4. Threats to Validity

CHAPTER 5

Conclusions

This chapter summarises the contributions, revisits the research questions, and outlines directions for future work.

5.1. Summary

5.2. Revisiting the Research Questions

5.3. Contributions

5.4. Limitations

5.5. Future Work

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APPENDIX A

Appendix Title

A.1. Section in the Appendix